Optimal degree bounds for sum-of-squares representations

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Abstract

The moment-sum-of-squares (SOS) hierarchy provides a sequence of bounds on the minimum of a polynomial over a basic closed semialgebraic set, i.e., a subset of $\mathbb{R}^n$ defined by finitely many polynomial constraints. We establish optimal bounds on the error of the SOS hierarchy in the following settings:

1. If the constraints define an Archimedean quadratic module: the error of the Putinar-based hierarchy at level k is $\Theta((\log k)^4/k^2)$ in the worst case.
2. If the constraints define a compact semialgebraic set: the error of the Schmüdgen-based hierarchy at level k is $\Theta((\log k)^2/k^2)$ in the worst case.
3. If the constraints define an Archimedean quadratic module and satisfy linear independence constraint qualification conditions (LICQ): the error of the hierarchy at level k is $\Theta(1/k^4)$ in the worst case. Here, the upper bound holds already for the Putinar-based hierarchy, while the lower bound holds even against the Schmüdgen-based hierarchy.

Equivalently, these give best-possible effective versions of Putinar's and Schmüdgen's Positivstellensätze for these settings (in terms of their dependence on k).

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Statement

This is an announcement of results which were produced over the course of several months with extensive use of generative AI. Several subsets of the authors independently arrived at (parts of) these results over that time period either by prompting various models, by independent human thought or, in most cases, by a mixture of both. The purpose of this announcement is to have a common, comprehensive reference for a set of results which we believe to essentially settle a line of research going back more than 20 years.

For each of the results claimed below, we provide a certificate of correctness in Lean. These can be viewed at https://github.com/Lucas-Slot/OptimalConvergenceRates. All theorem statements marked with **Lean** have a direct formalization in that repository. This announcement does not contain proofs beyond these certificates. In particular, we have opted *not* to publish any proofs generated directly by AI. Our motivation for omitting such proofs is twofold:

- They are not suited towards furthering human understanding of the underlying math;
- They do not properly incorporate, reference, or credit existing ideas from the literature.

The authors are preparing expositions of all claimed results that take into account the two points above. We intend to provide these in a future version of this preprint. *We would like to do so as a community, and we welcome contributions from others who may have been working on these or similar problems.*

1 Introduction

Let $\mathbb{R}[\mathbf{x}]$ denote the polynomial ring in n variables $\mathbf{x} = (x_1, x_2, \dots, x_n)$. For a tuple of polynomials $\mathbf{g} = (g_1, g_2, \dots, g_m)$, we write $\mathcal{S}(\mathbf{g})$ for the (basic, closed) semialgebraic set

$$\mathcal{S}(\mathbf{g}) := \{\mathbf{x} \in \mathbb{R}^n : g_1(\mathbf{x}) \geq 0, \dots, g_m(\mathbf{x}) \geq 0\}.$$

We consider the classical problem of certifying whether a given polynomial $p \in \mathbb{R}[\mathbf{x}]$ belongs to the cone $\mathcal{P}(\mathbf{g})$ of nonnegative polynomials on $\mathcal{S}(\mathbf{g})$. That is, whether $p(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathcal{S}(\mathbf{g})$. One way to certify *global* nonnegativity of p is to write it as a *sum of squares*, i.e.,

$$p(\mathbf{x}) = q_1(\mathbf{x})^2 + \ldots + q_\ell(\mathbf{x})^2 \quad (q_i \in \mathbb{R}[\mathbf{x}]).$$

We write $\Sigma[\mathbf{x}] \subseteq \mathbb{R}[\mathbf{x}]$ for the cone of all polynomials admitting such a decomposition. This idea can be extended to certify nonnegativity over $\mathcal{S}(\mathbf{g})$ as follows. For $k \in \mathbb{N}$, consider the (truncated) quadratic module and preordering associated to $\mathbf{g}$, given by

$$\mathcal{Q}(\mathbf{g})_k := \left\{ \sum_{i=0}^{m} g_i \sigma_i \,:\, \sigma_i \in \Sigma[\mathbf{x}] \text{ and } \deg(g_i\sigma_i) \leq 2k \text{ for all } 0 \leq i \leq m \right\}, \tag{1}$$

$$\mathcal{T}(\mathbf{g})_k := \left\{ \sum_{I \subseteq [m]} g_I \sigma_I \,:\, \sigma_I \in \Sigma[\mathbf{x}] \text{ and } \deg(g_I\sigma_I) \leq 2k \text{ for all } I \subseteq [m] \right\}. \tag{2}$$

Here, $g_I := \prod_{i \in I} g_i$ for $I \subseteq [m] = \{1, 2, \dots, m\}$, and we have adopted the notational convention that $g_0 = g_\emptyset = 1$. We write $\mathcal{Q}(\mathbf{g}) = \bigcup_{k=0}^{\infty} \mathcal{Q}(\mathbf{g})_k$ and $\mathcal{T}(\mathbf{g}) = \bigcup_{k=0}^{\infty} \mathcal{T}(\mathbf{g})_k$ for the degree-unrestricted versions, respectively. It follows immediately from the definition that

$$\mathcal{Q}(\mathbf{g})_k \subseteq \mathcal{T}(\mathbf{g})_k \subseteq \mathcal{P}(\mathbf{g}) \quad (\forall k \in \mathbb{N}). \tag{3}$$

Krivine [Kri64] and Stengle [Ste74] showed that any nonnegative polynomial on $\mathcal{S}(\mathbf{g})$ can be represented as a *ratio* of polynomials in the preordering. This generalizes Artin's solution to Hilbert's seventeenth problem [Art27]. A natural question is then whether all nonnegative polynomials on $\mathcal{S}(\mathbf{g})$ lie in $\mathcal{T}(\mathbf{g})$ or even in $\mathcal{Q}(\mathbf{g})$. This turns out not to be true in general. However, the classical *Positivstellensätze* of Putinar and Schmüdgen tell us that the inclusions (3) are "asymptotically tight" for (algebraically) compact (basic) semialgebraic sets, in the following sense:

Theorem 1.1 (Putinar's Positivstellensatz [Put93])**.** *Suppose that $\mathcal{Q}(\mathbf{g})$ is Archimedean.*[1] *Let $p \in \mathcal{P}(\mathbf{g})$. For every $\varepsilon > 0$, there exists $k \in \mathbb{N}$ such that $p + \varepsilon \in \mathcal{Q}(\mathbf{g})_k$.*

Theorem 1.2 (Schmüdgen's Positivstellensatz [Sch91])**.** *Suppose that $\mathcal{S}(\mathbf{g}) \subseteq \mathbb{R}^n$ is compact. Let $p \in \mathcal{P}(\mathbf{g})$. For every $\varepsilon > 0$, there exists $k \in \mathbb{N}$ such that $p + \varepsilon \in \mathcal{T}(\mathbf{g})_k$.*

In particular, these imply that any polynomial *strictly* positive on $\mathcal{S}(\mathbf{g})$ lies in $\mathcal{Q}(\mathbf{g})$ and $\mathcal{T}(\mathbf{g})$.

1.1 Effective Positivstellensätze and convergence of the SOS hierarchy.

There has been substantial interest in proving (asymptotic) bounds on the degree $k = k(\varepsilon)$ required in the above theorems as $\varepsilon \to 0$. This is motivated in large part by the fact that such bounds translate directly to convergence rates for the celebrated moment-SOS hierarchy for polynomial optimization [Las01, Par00]. This can be seen as follows: For a nonempty compact feasible set $\mathcal{S}(\mathbf{g})$ and an objective $f \in \mathbb{R}[\mathbf{x}]$, put $p = f - f_{\min}$, where $f_{\min} := \min_{\mathbf{x} \in \mathcal{S}(\mathbf{g})} f(\mathbf{x})$. Note that $p \in \mathcal{P}(\mathbf{g})$ and has zeros in $\mathcal{S}(\mathbf{g})$. For $\mathcal{C} \in \{\mathcal{Q}(\mathbf{g}), \mathcal{T}(\mathbf{g})\}$, define

$$f_{k,\mathcal{C}} := \sup\{\lambda \in \mathbb{R} : f - \lambda \in \mathcal{C}_k\}, \tag{4}$$

$$\varepsilon_{p,k,\mathcal{C}} := \inf\{\varepsilon \geq 0 : p + \varepsilon \in \mathcal{C}_k\} = f_{\min} - f_{k,\mathcal{C}}. \tag{5}$$

The parameter $\varepsilon_{p,k,\mathcal{C}}$ is called the *error* of the moment-SOS hierarchy at level k. Its asymptotic behavior as $k \to \infty$ is referred to as the *convergence rate* of the hierarchy. Equivalently, in the language of Theorems 1.1 and 1.2, $\varepsilon_{p,k,\mathcal{C}}$ corresponds to a degree requirement

$$k_{p,\varepsilon,\mathcal{C}} := \min\{k \in \mathbb{N} : p + \varepsilon \in \mathcal{C}_k\}. \tag{6}$$

The central question we address in this paper is the following:

Question 1.3. Under suitable assumptions on $\mathbf{g}$, what is the dependence of $\varepsilon_{p,k,\mathcal{C}}$ on k as $k \to \infty$ (or, equivalently, of $k_{p,\varepsilon,\mathcal{C}}$ on ε as $\varepsilon \to 0$)?

This question has been well-studied in the literature. Below, we give a brief overview of the current state of the art. We end the overview with Table 1, which compares the state of the art to the new results provided by this work. We present our results in detail in Section 2. Finally, we give a (hopefully) complete history of the development in Section 3 for reference.

1.2 State of the art

1.2.1 Upper degree bounds and convergence rates.

Upper bounds on the error $\varepsilon_{p,k,\mathcal{C}}$, or equivalently, on the required degree $k_{p,\varepsilon,\mathcal{C}}$ provide a safety guarantee that the hierarchy cannot converge slower than a stated rate, even on the hardest instances in the class.

[1] Meaning that $\mathcal{Q}(\mathbf{g})$ contains the ball constraint $N - \|\mathbf{x}\|^2$ for some $N > 0$. In particular, $\mathcal{S}(\mathbf{g})$ is then compact.

The literature on these upper bounds can be roughly divided into two streams. On the one hand, there is a line of works establishing such bounds under general constraint sets $\mathbf{g}$. Roughly speaking, this has culminated in the following: For every $\mathbf{g}$ defining an Archimedean quadratic module, there is a constant $c(\mathbf{g}) > 0$, so that, for every $p \in \mathcal{P}(\mathbf{g})$,

$$\varepsilon_{p,k,\mathcal{Q}(\mathbf{g})} \leq O(k^{-1/c(\mathbf{g})}).$$

Here, the O-notation hides constants depending only on $p, \mathbf{g}$.[2] The same is true for $\mathcal{T}(\mathbf{g})$ under the weaker assumption that $\mathcal{S}(\mathbf{g})$ is compact (but with a potentially larger constant $c(\mathbf{g})$). Thus, the dependence of $\varepsilon_{p,k,\mathcal{C}}$ on k is at most inverse polynomial.

A long line of work has led to the following understanding of this inverse-polynomial dependence: The constant $c(\mathbf{g})$ can be bounded in terms of the so-called Łojasiewicz exponent $Ł(\mathbf{g})$ of $\mathbf{g}$ (Definition 2.20). For Putinar ($\mathcal{C} = \mathcal{Q}(\mathbf{g})$), one has $c(\mathbf{g}) \leq 2 \cdot Ł(\mathbf{g})$, and for Schmüdgen ($\mathcal{C} = \mathcal{T}(\mathbf{g})$), one has $c(\mathbf{g}) \leq Ł(\mathbf{g})$ [HKVZ26, TT26b]. In the best case (when $\mathbf{g}$ satisfies a type of constraint qualification, see Definition 2.1), we have $Ł(\mathbf{g}) = 1$, leading to $c(\mathbf{g}) = 2$ and $c(\mathbf{g}) = 1$, respectively.

On the other hand, there is a series of works investigating error bounds for specific sets of constraints $\mathbf{g}$, corresponding to fundamental domains of optimization. The most well-studied examples are

$$\mathbf{g}_{\text{sph}} = (1 - \|\mathbf{x}\|^2, \|\mathbf{x}\|^2 - 1), \text{ defining the unit sphere } \mathcal{S}(\mathbf{g}) = S^{n-1}; \tag{7}$$

$$\mathbf{g}_{\text{cube}} = (1 - x_i^2 : 1 \leq i \leq n), \text{ defining the unit hypercube } \mathcal{S}(\mathbf{g}) = [-1, 1]^n. \tag{8}$$

For these examples, one can prove much better degree bounds than the general results above would predict. In particular, it is known that $c(\mathbf{g}_{\text{cube}}) = c(\mathbf{g}_{\text{sph}}) = 1/2$ suffice for certificates in the preordering [FF21, Slo22]. (Note that, for the standard description of the unit sphere, the quadratic module and preordering coincide.) For representations in the quadratic module of the hypercube, one can select $c(\mathbf{g}_{\text{cube}}) = 1/2$, up to polylogarithmic factors.

1.2.2 Lower degree bounds and convergence obstructions.

Much less is known about *lower bounds* on $\varepsilon_{p,k,\mathcal{C}}$ in terms of k (or, equivalently, on $k_{p,\varepsilon,\mathcal{C}}$ in terms of ε). Such lower bounds represent necessity statements: they quantify intrinsic limitations of the method (or of the representation class), and rule out uniformly better guarantees without changing assumptions or algorithms. Essentially, only two lower bound constructions are known.

First, a classical example due to Stengle [Ste96] shows that degree $k = \Omega(\varepsilon^{-1/2})$ is needed to represent $1 - x^2 + \varepsilon$ in the preordering generated by $(1 - x^2)^3$. That is, $c(\mathbf{g}) \geq 1/2$ for that example. Recently, Henrion [Hen26] showed that this analysis is tight, and derived an exact relaxation order. Stengle's construction relies fundamentally on the non-standard choice of representation for the interval. Indeed, any polynomial p nonnegative on $[-1, 1]$ admits a certificate in $\mathcal{T}(1 - x^2)$ of degree at most $\deg(p) + 1$ (by the Markov–Lukács theorem). The key point is that the constraint $(1 - x^2)^3$ is degenerate at the boundary of $[-1, 1]$ (cf. Definition 2.1 below).

Second, for semialgebraic sets with nondegenerate description, to our knowledge, the only available lower bound is due to [BS24]. There, the authors show that $(1 - x_1^2)(1 - x_2^2) + \varepsilon$ is not an element of the quadratic module $\mathcal{Q}(1 - x_1^2, 1 - x_2^2)_k$ unless $k = \Omega(\varepsilon^{-1/8})$. No lower bounds are available for the preordering of the unit hypercube, or for the quadratic module of the unit sphere (with standard description).

[2] In many of the works discussed below, these constants are made (more or less) explicit. For simplicity, we do not specify them in this overview.

1.3 Comparison to new results

In Table 1 below, we give an overview of the state-of-the-art results discussed above, and compare them to the new (optimal) results obtained in this work.

		Previous work		**This work**
Setting	Certificate	Upper bound $O(\cdot)$	Lower bound $\Omega(\cdot)$	**Optimal rate** $\Theta(\cdot)$
Compact	$\mathcal{T}(\mathbf{g})$	$k^{-1/\text{Ł}(\mathbf{g})}$	k^{-2}	$(\log k)^2/k^2$
Archimedean	$\mathcal{Q}(\mathbf{g})$	$k^{-1/(2\text{Ł}(\mathbf{g}))}$	k^{-2}	$(\log k)^4/k^2$
Hypercube	$\mathcal{T}(\mathbf{g}_{\text{cube}})$	k^{-2}	—	k^{-4}
Hypercube	$\mathcal{Q}(\mathbf{g}_{\text{cube}})$	$(\log k)^3/k^2$	k^{-8}	k^{-4}
Archimedean + $\text{Ł}(\mathbf{g}) = 1$	$\mathcal{Q}(\mathbf{g})$	$k^{-1/2}$	k^{-8}	k^{-2}
Archimedean + LICQ	$\mathcal{Q}(\mathbf{g})$	$k^{-1/2}$	k^{-8}	k^{-4}

Table 1: Comparison of prior upper and lower bounds on the asymptotic error $\varepsilon_{p,k,\mathcal{C}}$, as $k \to \infty$ with the optimal rates established in this work (Putinar = $\mathcal{Q}$, Schmüdgen = $\mathcal{T}$). Upper bounds hold for all admissible instances, whereas lower bounds are attained by specific instances. The rates in the last column are optimal in the worst-case sense up to multiplicative constants (that may depend on the dimension n as well as on p and $\mathbf{g}$).

2 Contributions

We provide a series of matching upper and lower degree bounds for Putinar's and Schmüdgen's Positivstellensatz in three regimes: (1) general constraint sets $\mathbf{g}$ defining an Archimedean quadratic module or compact semialgebraic set, respectively; (2) the standard description $\mathbf{g}_{\text{cube}}$ of the unit hypercube $[-1, 1]^n$; (3) general $\mathbf{g}$ satisfying the following constraint qualification condition.

Definition 2.1. We say that a tuple of polynomials $\mathbf{g}$ satisfies Linear Independence Constraint Qualification (LICQ) if, for every $\mathbf{x} \in \mathcal{S}(\mathbf{g})$, the gradients of the active constraints at $\mathbf{x}$, being

$$\big(\nabla g(\mathbf{x}) : g \in \mathbf{g},\ g(\mathbf{x}) = 0\big), \tag{9}$$

are linearly independent (in particular, nonzero). By convention, if the tuple $\mathbf{g}$ contains both g and $-g$, then only one of their gradients is considered in eq. (9).

Notable examples of tuples that satisfy the LICQ include $\mathbf{g}_{\text{sph}} = (1 - \|\mathbf{x}\|^2, \|\mathbf{x}\|^2 - 1)$ corresponding to the unit sphere, and $\mathbf{g}_{\text{ball}} = (1 - \|\mathbf{x}\|^2)$ corresponding to the unit ball.

2.1 Effective Putinar: Archimedean quadratic modules

The following theorems provide an asymptotically optimal effective version of Putinar's Positivstellensatz (Theorem 1.1). Namely, they establish that degree $k = \Theta(\log(1/\varepsilon)^2/\sqrt{\varepsilon})$ is sufficient in general, and necessary in the worst case. We first show the upper degree bound.

Theorem 2.2 (Lean | Archimedean, upper bounds on degree and error)**.** *Let* $\mathbf{g}$ *be a set of constraints defining an Archimedean quadratic module* $\mathcal{Q}(\mathbf{g})$*, and a nonempty semialgebraic set* $\mathcal{S}(\mathbf{g})$*. Let* $p \in \mathcal{P}(\mathbf{g})$*. Then, there exist constants* $C, k_0, \varepsilon_0 > 0$*, depending only on* p *and* $\mathbf{g}$*, such that*

$$\varepsilon_{p,k,\mathcal{Q}(\mathbf{g})} \leq C \cdot \frac{(\log k)^4}{k^2} \quad (\forall k \geq k_0); \quad k_{p,\varepsilon,\mathcal{Q}(\mathbf{g})} \leq C \cdot \frac{\log(1/\varepsilon)^2}{\sqrt{\varepsilon}} \quad (\forall \varepsilon \leq \varepsilon_0).$$

The lower degree bound uses the following instance.

Example 2.3. Let $n = 3$ and consider the constraint set

$$\mathbf{g} = (x_1, x_2,\ x_1x_2x_3 + x_3^5,\ 1 - \|\mathbf{x}\|^2).$$

The quadratic module $\mathcal{Q}(\mathbf{g})$ is Archimedean. The semialgebraic set is $\mathcal{S}(\mathbf{g}) = \{\mathbf{x} : \mathbf{x} \geq 0,\ \|\mathbf{x}\|^2 \leq 1\}$, i.e., the intersection of the unit ball with the nonnegative orthant. To see this, note that given the first two constraints, the third is equivalent to $x_3 \geq 0$. In particular, $\mathcal{S}(\mathbf{g})$ is nonempty, and thus $\mathbf{g}$ satisfies the conditions of Theorem 2.2.

Theorem 2.4 (Lean | Archimedean, lower bounds on degree and error)**.** *Let* $n = 3$*, and let* $\mathbf{g}$ *be as in Example 2.3. Consider the polynomial* $p = x_3$*, which is nonnegative on* $\mathcal{S}(\mathbf{g})$*. There exist constants* $c, k_0, \varepsilon_0 > 0$ *such that*

$$\varepsilon_{p,k,\mathcal{Q}(\mathbf{g})} \geq c \cdot \frac{(\log k)^4}{k^2} \quad (\forall k \geq k_0); \quad k_{p,\varepsilon,\mathcal{Q}(\mathbf{g})} \geq c \cdot \frac{\log(1/\varepsilon)^2}{\sqrt{\varepsilon}} \quad (\forall \varepsilon \leq \varepsilon_0); \tag{10}$$

Remark 2.5. The constraint set $\mathbf{g}$ of Example 2.3 does not satisfy LICQ. Indeed, for $g_3(x_1, x_2, x_3) = x_1x_2x_3 + x_3^5$, we have $g_3(0,0,0) = 0$ and $\nabla g_3(0,0,0) = (0,0,0)$.

2.2 Effective Schmüdgen: Compact semialgebraic sets

The following theorems provide an asymptotically optimal effective version of Schmüdgen's Positivstellensatz (Theorem 1.2). Namely, they establish that degree $k = \Theta(\log(1/\varepsilon)/\sqrt{\varepsilon})$ is sufficient in general, and necessary in the worst case. (This result differs from that of the previous section by a factor $\log(1/\varepsilon)$).

Theorem 2.6 (Lean | Compact, upper bounds on degree and error)**.** *Let* $\mathbf{g}$ *be a set of constraints defining a compact, nonempty semialgebraic set* $\mathcal{S}(\mathbf{g})$*. Let* $p \in \mathcal{P}(\mathbf{g})$*. Then, there exist constants* $C, k_0, \varepsilon_0 > 0$*, depending only on* p *and* $\mathbf{g}$*, such that*

$$\varepsilon_{p,k,\mathcal{T}(\mathbf{g})} \leq C \cdot \frac{(\log k)^2}{k^2} \quad (\forall k \geq k_0); \quad k_{p,\varepsilon,\mathcal{T}(\mathbf{g})} \leq C \cdot \frac{\log(1/\varepsilon)}{\sqrt{\varepsilon}} \quad (\forall \varepsilon \leq \varepsilon_0).$$

The lower bound is based on the following bivariate example.

Example 2.7. Let $n = 2$ and consider the constraint set

$$\mathbf{g} = (x_2(x_1^2 + x_2^2),\ 1 - x_1^2 - x_2^2).$$

The semialgebraic set $\mathcal{S}(\mathbf{g}) = \{x_2 \geq 0,\ \|\mathbf{x}\|^2 \leq 1\}$ is compact. In fact, $\mathcal{Q}(\mathbf{g})$ is Archimedean.

Theorem 2.8 (Lean | Compact, lower bounds on degree and error)**.** *Let* $n = 2$*, and let* $\mathbf{g}$ *as in Example 2.7. Consider the polynomial* $p = x_2$*, which is nonnegative on* $\mathcal{S}(\mathbf{g})$*. There exist constants* $c, k_0, \varepsilon_0 > 0$ *such that*

$$\varepsilon_{p,k,\mathcal{T}(\mathbf{g})} \geq c \cdot \frac{(\log k)^2}{k^2} \quad (\forall k \geq k_0); \quad k_{p,\varepsilon,\mathcal{T}(\mathbf{g})} \geq c \cdot \frac{\log(1/\varepsilon)}{\sqrt{\varepsilon}} \quad (\forall \varepsilon \leq \varepsilon_0).$$

Remark 2.9. The constraint set $\mathbf{g}$ of Example 2.7 does not satisfy the LICQ (Definition 2.1). Indeed, for $g_1(x_1, x_2) = x_2(x_1^2 + x_2^2)$, we have $g_1(0,0) = 0$ and $\nabla g_1(0,0) = (0,0)$.

2.3 The unit hypercube

In this section, we consider the unit hypercube $[-1,1]^n$, with its standard description

$$[-1,1]^n = \mathcal{S}(\mathbf{g}_{\text{cube}}), \qquad \text{where } \mathbf{g}_{\text{cube}} = (1 - x_i^2 \,:\, 1 \le i \le n).$$

The following theorems provide asymptotically optimal effective versions of both Putinar's and Schmüdgen's Positivstellensätze as applied to the unit hypercube (with its standard description). Namely, they establish for both that degree $k = \Theta(1/\sqrt[4]{\varepsilon})$ is sufficient in general, and necessary in the worst case. For Putinar, the worst-case example uses $n = 2$ variables. For Schmüdgen, the worst-case example uses $n = 4$ variables.

Theorem 2.10 (**Lean** | Hypercube, upper bounds on degree and error)**.** *For every $n \ge 1$ and $d \ge 0$, there exist constants $C_{n,d}, k_{n,d}, \varepsilon_{n,d} > 0$, depending only on n and d, such that, for every polynomial $p \in \mathcal{P}(\mathbf{g}_{\text{cube}})$ of degree at most d,*

$$\varepsilon_{p,k,\mathcal{Q}(\mathbf{g}_{\text{cube}})} \le C_{n,d} \cdot \frac{\|p\|_\infty}{k^4} \quad (\forall k \ge k_{n,d}); \qquad k_{p,\varepsilon,\mathcal{Q}(\mathbf{g}_{\text{cube}})} \le C_{n,d} \cdot \left(\sqrt[4]{\frac{\|p\|_\infty}{\varepsilon}} + 1 \right) \quad (\forall \varepsilon \le \varepsilon_{n,d}).$$

Here, $\|p\|_\infty := \max_{\mathbf{x} \in [-1,1]^n} |p(\mathbf{x})|$. As a consequence, the same holds for $\mathcal{T}(\mathbf{g}_{\text{cube}})$.

We first present the instance used in the lower degree bound for Putinar-type certificates, for $n = 2$.

Example 2.11. Let $n = 2$. Consider the polynomial

$$\mathrm{BS}(x_1, x_2) := (1 - x_1^2)(1 - x_2^2).$$

This polynomial is nonnegative on $[-1,1]^2$. It is an element of $\mathcal{T}(\mathbf{g}_{\text{cube}})$ by definition, but it is not an element of $\mathcal{Q}(\mathbf{g}_{\text{cube}})$, see [BS24].

Theorem 2.12 (**Lean** | Hypercube, lower bounds on degree and error I)**.** *Let $n = 2$, and let $p = \mathrm{BS} \in \mathbb{R}[\mathbf{x}]$ be as in Example 2.11. There exist absolute constants $c, k_0, \varepsilon_0 > 0$ such that,*

$$\varepsilon_{p,k,\mathcal{Q}(\mathbf{g}_{\text{cube}})} \ge c \cdot \frac{1}{k^4} \quad (\forall k \ge k_0); \qquad k_{p,\varepsilon,\mathcal{Q}(\mathbf{g}_{\text{cube}})} \ge c \cdot \frac{1}{\sqrt[4]{\varepsilon}} \quad (\forall \varepsilon \le \varepsilon_0).$$

These bounds are asymptotically tight in light of Theorem 2.10.

Remark 2.13. The statement of Theorem 2.12 was conjectured based on experiments with a high-precision SDP solver [BK26]. In fact, Baldi and Kočvara conjecture it holds more generally for the representation of $g_1 \cdot g_2 + \varepsilon$ in $\mathcal{Q}(g_1, g_2)$, where g_1, g_2 are arbitrary polynomials that intersect transversally in $\mathbb{R}^n$ and generate an Archimedean quadratic module.

We now present two instances that lead to lower degree bounds for Schmüdgen-type certificates for $n = 3$, and $n \ge 4$.

Example 2.14. Let $n = 3$. Consider the Motzkin form

$$\mathrm{Mtz}(x_1, x_2, x_3) := x_1^4 x_2^2 + x_1^2 x_2^4 + x_3^6 - 3 x_1^2 x_2^2 x_3^2.$$

The polynomial Mtz is globally nonnegative, but it is not a sum of squares, see [Rez07]. In fact, it is not an element of $\mathcal{T}(\mathbf{g}_{\text{cube}})$.

Theorem 2.15 (Lean | Hypercube, lower bounds on degree and error II)**.** *Let $n = 3$, and let $p = \mathrm{Mtz} \in \mathbb{R}[\mathbf{x}]$ be as in Example 2.14. There exist absolute constants $C, c, k_0, \varepsilon_0 > 0$ such that*

$$C \cdot \frac{1}{k^6} \geq \varepsilon_{p,k,\mathcal{T}(\mathbf{g}_{\mathrm{cube}})} \geq c \cdot \frac{1}{k^6} \quad (\forall k \geq k_0); \quad C \cdot \frac{1}{\sqrt[6]{\varepsilon}} \geq k_{p,\varepsilon,\mathcal{T}(\mathbf{g}_{\mathrm{cube}})} \geq c \cdot \frac{1}{\sqrt[6]{\varepsilon}} \quad (\forall \varepsilon \leq \varepsilon_0).$$

Example 2.16. Let $n = 4$. Consider the Choi–Lam form

$$\mathrm{CL}(x_1, x_2, x_3, x_4) := x_1^4 + x_2^2 x_3^2 + x_3^2 x_4^2 + x_4^2 x_2^2 - 4 x_1 x_2 x_3 x_4.$$

The polynomial CL is globally nonnegative, but is not a sum of squares, see [Rez07]. In fact, it is not an element of $\mathcal{T}(\mathbf{g}_{\mathrm{cube}})$.

Theorem 2.17 (Lean | Hypercube, lower bounds on degree and error III)**.** *Let $n = 4$, and let $p = \mathrm{CL} \in \mathbb{R}[\mathbf{x}]$ be as in Example 2.16. There exist absolute constants $c, k_0, \varepsilon_0 > 0$ such that,*

$$\varepsilon_{p,k,\mathcal{T}(\mathbf{g}_{\mathrm{cube}})} \geq c \cdot \frac{1}{k^4} \quad (\forall k \geq k_0); \quad k_{p,\varepsilon,\mathcal{T}(\mathbf{g}_{\mathrm{cube}})} \geq c \cdot \frac{1}{\sqrt[4]{\varepsilon}} \quad (\forall \varepsilon \leq \varepsilon_0).$$

Consequently, the same holds for $\mathcal{Q}(\mathbf{g}_{\mathrm{cube}})$. These bounds are asymptotically tight in light of Theorem 2.10.

Remark 2.18. We presented three different lower bounds, respectively for $n = 2$, $n = 3$, $n = 4$.

- For $n = 2$, we get a lower bound only w.r.t. the quadratic module. We cannot hope for more, because the preordering is *saturated* there, i.e., $\mathcal{T}(\mathbf{g}_{\mathrm{cube}}) = \mathcal{P}(\mathbf{g}_{\mathrm{cube}})$ for $n = 2$ [Sch06].
- For $n = 3$, $n = 4$, we get lower bounds by taking a globally nonnegative (but not SOS) polynomial, and homogenizing it. The strength of the bound is better if the degree of the polynomial is lower. Note that this follows the Hilbert cases for homogeneous polynomials: at degree 2, all globally nonnegative forms are SOS. For $n = 3$, this is true up to and including degree 4. For $n \geq 4$, there are globally nonnegative forms of degree 4 that are not SOS.

2.4 Transfer theorems, constraint qualifications and Łojasiewicz exponents

Given the convergence rate results on the hypercube, a natural question is whether such rates also hold for other "simple" sets such as the unit sphere, unit ball, and standard simplex. Rather than providing separate statements for each of these sets, we observe that all of their standard descriptions give rise to Archimedean quadratic modules, and satisfy the constraint qualification LICQ. The following Theorem 2.19 shows that this is sufficient to obtain an $O(k^{-4})$ rate of convergence. This theorem thus implies a rate of $O(k^{-4})$ for standard descriptions of the unit ball, the unit sphere and the simplex. The proof of this theorem relies crucially on the $O(k^{-4})$ convergence rate of the Putinar-based hierarchy on the hypercube stated in Theorem 2.10. Theorem 2.19 can be viewed as transferring that result to any Archimedean quadratic module for which LICQ holds. We will provide a mathematical exposition of the proof of this result in a forthcoming version of this preprint. For completeness, note that the instance used in the lower degree bound for general Archimedean quadratic modules (Theorem 2.4) does not satisfy LICQ.

Theorem 2.19 (Lean)**.** *Let $\mathbf{g}$ be a set of constraints defining an Archimedean quadratic module. Suppose that LICQ holds for $\mathbf{g}$ (Definition 2.1), and that $\mathcal{S}(\mathbf{g})$ is nonempty. Let $p \in \mathcal{P}(\mathbf{g})$. Then, there exist constants $C, k_0, \varepsilon_0 > 0$, depending only on p and $\mathbf{g}$, such that*

$$\varepsilon_{p,k,\mathcal{Q}(\mathbf{g})} \leq C \cdot \frac{1}{k^4} \quad (\forall k \geq k_0); \quad k_{p,\varepsilon,\mathcal{Q}(\mathbf{g})} \leq C \cdot \frac{1}{\sqrt[4]{\varepsilon}} \quad (\forall \varepsilon \leq \varepsilon_0).$$

The convergence rate of the SOS hierarchy for a constraint set $\mathbf{g}$ has been connected to the Łojasiewicz exponent $Ł(\mathbf{g})$ of $\mathbf{g}$ at several places in the literature [Sch04, BM23, BMP25, HKVZ26]. It is known that if $\mathbf{g}$ satisfies LICQ, we have $Ł(\mathbf{g}) = 1$ [BM23]. It is thus a reasonable question whether the quartic convergence established above might extend to all (Archimedean) $\mathbf{g}$ with $Ł(\mathbf{g}) = 1$. This turns out to be false, as the following example shows.

Definition 2.20. Let $\mathbf{g}$ be a set of polynomial constraints defining a compact, nonempty semialgebraic set $\mathcal{S}(\mathbf{g})$. We write $Ł(\mathbf{g})$ for the Łojasiewicz exponent of $\mathbf{g}$, which is the smallest $Ł \geq 1$ so that the following holds: For every closed, bounded neighborhood $\mathcal{N} \supseteq \mathcal{S}(\mathbf{g})$, there exists a $C_{\mathcal{N}} > 0$ with

$$\operatorname{dist}(\mathbf{x}, \mathcal{S}(\mathbf{g}))^{Ł} \leq C_{\mathcal{N}} \cdot \max\{0, \max_{g \in \mathbf{g}} -g(\mathbf{x})\} \quad \forall \mathbf{x} \in \mathcal{N}.$$

Proposition 2.21 (see, e.g., [BMP25])**.** *Let $\mathbf{g}$ be a set of polynomial constraints. Assume that $\mathcal{Q}(\mathbf{g})$ is Archimedean and that $\mathbf{g}$ satisfies LICQ (Definition 2.1). Then $Ł(\mathbf{g}) = 1$.*

Example 2.22. Let $n = 3$ and write

$$\begin{aligned} q_1(x_1, x_2) &= x_1^2 - 3x_2^2, & q_2(x_1, x_2) &= x_2^2 - 3x_1^2, \\ q_3(x_1, x_2) &= 4x_1x_2 - x_1^2 - x_2^2, & q_4(x_1, x_2) &= -4x_1x_2 - x_1^2 - x_2^2. \end{aligned}$$

Consider the constraint set $\mathbf{g} = (g_i : 1 \leq i \leq 5)$ given by

$$g_i = x_3 - q_i \quad (1 \leq i \leq 4), \quad g_5 = 1 - x_1^2 - x_2^2 - x_3^2.$$

The quadratic module $\mathcal{Q}(\mathbf{g})$ is Archimedean. The Łojasiewicz exponent of $\mathbf{g}$ is $Ł(\mathbf{g}) = 1$. The LICQ for $\mathbf{g}$ fails at $\mathbf{x} = (0, 0, 0)$. We have $x_3 \geq 0$ for all $\mathbf{x} \in \mathcal{S}(\mathbf{g})$.

Theorem 2.23 (**Lean**)**.** *Let $n = 3$ and consider the constraint set $\mathbf{g}$ of Example 2.22. Let $p = x_3$. Then $p \in \mathcal{P}(\mathbf{g})$. There exist constants $c, k_0, \varepsilon_0 > 0$ such that,*

$$\varepsilon_{p,k,\mathcal{Q}(\mathbf{g})} \geq c \cdot \frac{1}{k^2} \quad (\forall k \geq k_0); \quad k_{p,\varepsilon,\mathcal{Q}(\mathbf{g})} \geq c \cdot \frac{1}{\sqrt{\varepsilon}} \quad (\forall \varepsilon \leq \varepsilon_0).$$

Remark 2.24. By combining Theorem 2.10 with the techniques presented in [HKVZ26], it follows that for any set of constraints $\mathbf{g}$ defining an Archimedean quadratic module $\mathcal{Q}(\mathbf{g})$ and a nonempty semialgebraic set $\mathcal{S}(\mathbf{g})$, and any $p \in \mathcal{P}(\mathbf{g})$, there exist constants $C, k_0, \varepsilon_0 > 0$, depending only on p and $\mathbf{g}$, such that

$$\varepsilon_{p,k,\mathcal{Q}(\mathbf{g})} \leq C \cdot \frac{1}{k^{2/Ł(\mathbf{g})}} \quad (\forall k \geq k_0); \quad k_{p,\varepsilon,\mathcal{Q}(\mathbf{g})} \leq C \cdot \frac{1}{\varepsilon^{Ł(\mathbf{g})/2}} \quad (\forall \varepsilon \leq \varepsilon_0).$$

In particular, a quadratic convergence-rate upper bound is obtained when $Ł(\mathbf{g}) = 1$.

3 Historical development of convergence rates and degree bounds

The following tables attempt to exhaustively survey the existing literature on Question 1.3. Each rate displayed below corresponds to $\varepsilon_{p,k,\mathcal{C}}$ as defined in (5), with the cone $\mathcal{C}$ specified in each row. In the first four tables, the "Rate" column displays upper bounds as $k \to \infty$, and, unless a row states otherwise, objectives are arbitrary polynomials; the stated bound holds up to constants which may depend on $\mathbf{g}$ and p (as noted earlier, many of the cited works make these constants more explicit). In the last table, the "Rate" column displays either lower bounds with the "Ω" notation or exact error

bounds. In Table 3, hypercube rows use the standard quadratic generators, or the corresponding standard interval description under a bounded degree conversion. In Table 4, for the ball, one has $\mathcal{Q} = \mathcal{T}$ because there is only one inequality generator. For any compact univariate basic closed set, one also has $\mathcal{Q} = \mathcal{T}$. In Table 5, the sparse rows assume variable cliques satisfying the so-called *running intersection property*; the objective is a sum of polynomials supported on individual cliques, and each constraint and certificate term is supported on one such clique.

In Table 6, the univariate examples use their displayed nonstandard generators. Their lower bounds do not apply to the interval described by $1 - x^2$. The Baldi–Slot objective [BS24, Thm. 4, Cor. 16] is already certified in $\mathcal{T}(\mathbf{g}_{\mathrm{cube}})_2$. The final row concerns SOS approximation and is not a lower bound for the hypercube hierarchy.

Table 2: General compact and Archimedean sets

Setting	**Certificate**	**Rate** $O(\cdot)$	**Note**	**Reference**
Compact $\mathcal{S}(\mathbf{g})$	$\mathcal{T}(\mathbf{g})$	$k^{-\alpha_{\mathbf{g}}}$	$\alpha_{\mathbf{g}} > 0$	[Sch04, Thms. 3–4]
Archimedean $\mathcal{Q}(\mathbf{g})$	$\mathcal{Q}(\mathbf{g})$	$(\log k)^{-\alpha_{\mathbf{g}}}$	$\alpha_{\mathbf{g}} > 0$	[NS07, Thm. 6]
Archimedean $\mathcal{Q}(\mathbf{g})$	$\mathcal{Q}(\mathbf{g})$	$k^{-2/(5n\text{Ł}(\mathbf{g}))}$	$n \geq 2$	[BM23, Thm. 1.7, Eq. (21)]
Archimedean $\mathcal{Q}(\mathbf{g})$	$\mathcal{Q}(\mathbf{g})$	$k^{-1/(7\text{Ł}(\mathbf{g})+3)}$		[BMP25, Cor. 3.3]
Archimedean $\mathcal{Q}(\mathbf{g})$	$\mathcal{Q}(\mathbf{g})$	$k^{-1/(2\text{Ł}(\mathbf{g}))}$		[HKVZ26, Thm. 4]
Compact $\mathcal{S}(\mathbf{g})$	$\mathcal{T}(\mathbf{g})$	$k^{-1/\text{Ł}(\mathbf{g})}$		[HKVZ26, Thm. 6]
Compact $\mathcal{S}(\mathbf{g})$	$\mathcal{T}(\mathbf{g})$	$k^{-1/\text{Ł}(\mathbf{g})}$	with normalization, via a subcone	[TT26b, Thm. 4.7, Cor. 4.8]
Compact $\mathcal{S}(\mathbf{g})$; with explicit hypercube generators	$\mathcal{Q}(\mathbf{g})$	$\dfrac{(\log k)^{3/(2\text{Ł}(\mathbf{g}))}}{k^{1/\text{Ł}(\mathbf{g})}}$		[TT26a, Thm. 3.7]

Table 3: Hypercube

Setting	**Certificate**	**Rate** $O(\cdot)$	**Note**	**Reference**
Hypercube	$\mathcal{T}(\mathbf{g}_{\mathrm{cube}})$	k^{-1}	via Handelman representation	[dKL10, Thm. 1.4]
Hypercube	$\mathcal{Q}(\mathbf{g}_{\mathrm{cube}})$	$(\log k)^{-1}$	*conditional* on a conjecture of de Klerk–Laurent.	[Mag15, Thm. 4]
Hypercube	$\mathcal{T}(\mathbf{g}_{\mathrm{cube}})$	k^{-2}		[LS23, Thm. 2]
Hypercube	$\mathcal{Q}(\mathbf{g}_{\mathrm{cube}})$	k^{-1}		[BS24, Thm. 3, Cor. 15]
Hypercube	$\mathcal{Q}(\mathbf{g}_{\mathrm{cube}})$	$(\log k)/k$	SOS approximation	[GdKV26a, Cor. 24]
Hypercube	$\mathcal{Q}(\mathbf{g}_{\mathrm{cube}})$	$(\log k)^3/k^2$	SOS approximation	[GdKV26b, Thm. 7]

Continued on the next page.

Table 3: Hypercube: convergence-rate upper bounds (continued)

Setting	**Certificate**	**Rate** $O(\cdot)$	**Note**	**Reference**
Hypercube	$\mathcal{T}(\mathbf{g}_{\mathrm{cube}})$	k^{-2}	trigonometric SOS	[BR23, Thm. 3.1]

Table 4: Sphere, ball, simplex, polytopes, and univariate sets

Setting	**Certificate**	**Rate** $O(\cdot)$	**Note**	**Reference**
Sphere	$\mathcal{Q}(\mathbf{g}_{\mathrm{sph}})$	k^{-1}	forms of even degree	[Rez95, Thm. 3.12]
Sphere	$\mathcal{Q}(\mathbf{g}_{\mathrm{sph}})$	k^{-1}	forms of even degree	[DW12, Cor. 7.2]
Sphere	$\mathcal{Q}(\mathbf{g}_{\mathrm{sph}})$	k^{-2}	forms of even degree $2d$; original theorem assumes $d \leq n$	[FF21, Thm. 2]
Sphere	$\mathcal{Q}(\mathbf{g}_{\mathrm{sph}})$	k^{-2}	forms of even degree; via moment-relaxation	[BL26, Thms. 3.1–3.2]
Sphere	$\mathcal{Q}(\mathbf{g}_{\mathrm{sph}})$	k^{-2}	extends [FF21] to arbitrary polynomials	[TT26b, Thm. 4.15, Cor. 4.16]
Unit ball	$\mathcal{Q} = \mathcal{T}$	k^{-2}		[Slo22, Thm. 3]
Simplex	$\mathcal{T}$	k^{-1}	Pólya-type representation	[PR01, Thm. 1]; [dKLP06, Thm. 1.1]
Simplex	LP-based approximation	k^{-1}	quadratic objectives	[BdK02, Thm. 3.2, Cor. 3.1]
Simplex	$\mathcal{T}$	k^{-2}		[Slo22, Thm. 4]
Compact polytope	$\mathcal{T}$	k^{-1}	linear inequality description	[TT26b, Thm. 4.14]
Compact univariate basic closed set	$\mathcal{Q} = \mathcal{T}$	k^{-2}	arbitrary given generators	[HSED26, Thm. 1]

Table 5: Products, liftings, affine transfers, and sparse constructions

Setting	**Certificate**	**Rate** $O(\cdot)$	**Note**	**Reference**
Product of two spheres	$\mathcal{T}$	k^{-2}		[Mag26, Thm. 3]
Any fixed finite product of spheres	$\mathcal{T}$	k^{-2}		[Mag26, Thm. 10]

Continued on the next page.

Table 5: Products, lifting, affine transfers, and sparse constructions (continued)

Setting	**Certificate**	**Rate $O(\cdot)$**	**Note**	**Reference**
Products of balls, simplices, hypercubes, and compatible mixtures	$\mathcal{T}$	k^{-2}		[Mag26, Cor. 12]; [TT26b, Thm. 3.3]
Two spheres	Matrix SOS, hence $\mathcal{T}$	k^{-2}	$f(x,y) = y^{\mathsf{T}} F(x) y$, with F a form of even degree	[FF21, Thm. 2]
Algebraic sections of products	$\mathcal{T}$	$k^{-1/Ł(\mathbf{g})}$	with suitable quadratic-rate kernels	[TT26b, Thm. 4.3]
Tensor products of compatible positive polynomial kernels	$\mathcal{T}$	Factor errors combine; slowest factor rate is retained		[Mag26, Thm. 11]
Positive lifting into a hypercube	Original $\mathcal{Q}$ or $\mathcal{T}$	Transfers hypercube degree bounds with controlled degree growth	followed by polynomial substitution	[HKVZ26, Thm. 2]
Hypercube	Sparse $\mathcal{T}(\mathbf{g}_{\mathrm{cube}})$	$k^{-2/(\omega+3)}$	ω is the largest variable-clique size	[KMRZ25, Thm. 6]
Block-Archimedean sparse problem	Sparse $\mathcal{Q}$	$k^{-\beta_{\mathrm{sp}}}$; if all $Ł_j = 1$: $k^{-18/(55\omega+238)}$	β_{sp} depends on clique sizes and $Ł_j$'s (local Łojasiewicz exponents)	[KMRZ25, Thm. 8]
Invertible affine images	Transported $\mathcal{Q}$ or $\mathcal{T}$	Same exponent	ellipsoids, parallelepipeds, affine simplices	Elementary degree-preserving affine substitution

Table 6: Lower bounds and exact-error constructions

Setting	**Certificate**	**Rate $\Omega(\cdot)$**	**Note**	**Reference**
$g = (1-x^2)^3$; $p = 1-x^2$	$\mathcal{Q}(g) = \mathcal{T}(g)$	k^{-2}		[Ste96, Thm. 4]
$g = (1-x^2)^3$; $p = 1-x^2$	$\mathcal{Q}(g) = \mathcal{T}(g)$	$\dfrac{1}{k(k-2)}$	$k \geq 3$, exact rate	[Hen26, Thm. 4]
$g = (1-x)^3(1+x)$; $p = 1-x$	$\mathcal{Q}(g) = \mathcal{T}(g)$	$\dfrac{1}{2k(k-1)}$	$k \geq 2$, exact rate	[HSED26, Thm. 2]
$\mathbf{g} = \{x_1^3 - x_2^2,\ 1-x_1\}$; $p = x_1$	$\mathcal{Q}(\mathbf{g})$	$\dfrac{1}{(2k-1)(2k-3)}$	$k \geq 2$, exact rate	[HSED26, Rmk. 3]

Continued on the next page.

Table 6: Lower bounds and exact-error constructions (continued)

Setting	**Certificate**	**Rate** $\Omega(\cdot)$	**Note**	**Reference**
Standard hypercube; $p = (1-x_1^2)(1-x_2^2)$	$\mathcal{Q}(\mathbf{g}_{\text{cube}})$	k^{-8}	$n \geq 2$;	[BS24, Thm. 4, Cor. 16]
Standard hypercube	Σ	k^{-2}	$n \geq 2$: SOS approximation	[GdKV26a, Thm. 10]

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